

Q5) A(vehicle) input is taken and a loop is used. Another inner loop is created. which is used to validate the input. Then depending on input. appropriate zones & vehicles are chosen and relevant counters are update. Finally a summary is generated.

PAC Chart

Given Data

Vehicles, User &
Vehicle category.

Required Results.

Total processed, accept
rejected, emergency

Processing

Data Validation, counters,
zones etc.

Alternative.

Nested
loops

IPO Chart.

Input.

Vehicles, type,
zone, permits.

Processing

Data Validation
Updating Counters

Output.

Validity

INPUT Vehicles

processed = Vehicles

accepted, rejected $\leftarrow 0$.

Cars, bikes, Vans = 0.

A $\leftarrow 20$, B $\leftarrow 40$, C $\leftarrow 15$

WHILE vehicles $\neq 0$:

data = FALSE.

WHILE data = FALSE:

INPUT category.

INPUT type

INPUT permit.

IF type $\neq$ 'C' AND type $\neq$ 'B'
AND type $\neq$ 'V':

PRINT 'Invalid Vehicle Category'

ELSE IF category $\neq$ 'F' AND category
 $\neq$ 'S' AND category $\neq$ 'G':

PRINT 'Invalid Category'

ELSE IF permit $\neq$ 'Y' AND permit
 $\neq$ 'N':

PRINT 'Invalid Permit'

ELSE:

IF permit = 'N':

INPUT Emergency.

IF Emergency ≠ 'Y' AND

Emergency ≠ 'N':

PRINT 'Invalid Emergency'.

ELSE:

Data = True.

ENDIF

ELSE:

Data = True.

ENDIF

ENDWHILE

IF permit = 'N' AND Emergency = 'N':

print ('Rejected').

rejected $\pm$ 1 = 1

ELSE.

IF category = 'F':

Zone = 'A'

ELSE IF category = 'S'

IF type = 'V'

Zone = 'C'

Date: _____

```
ELSE Zone = 'C'
ELSE Zone = 'B'
ELSE Zone = 'C'
ENDIF
```

```
IF Zone = 'A':
    IF type = 'C':
        IF A > 0:
            { PRINT ('Parket at A: Remaing', A)
              A = A - 1.
              Car += 1
              Accepted += 1
            ELSE
              PRINT 'Vehicle Rejected.'
              rejected += 1
            ENDIF
        ENDIF
```

```
IF type = 'B':
    IF A > 0:
        PRINT ('Parket at A: Remaing', A)
        A = A - 1.
        Bike += 1
        Accepted += 1
    ELSE
        PRINT ('Vehicle Rejected')
```

rejected += 1

IF type = 'E':

IF A > 1:

PRINT ('Successfully Parked at A', A)

A = A - 2.

Van += 1.

Accepted += 1.

ELSE:

PRINT ('Vehicle Rejected').

Rejected += 1.

ENDIF.

ENDIF

IF Zone = 'B':

IF Type = 'C':

IF B > 0:

PRINT ('Parked at B', B).

B = B - 1.

Car += 1.

Accepted += 1

ELSE: PRINT ('Vehicle Rejected').

Rejected += 1

ENDIF

IF type = 'B':

IF B > 0:

B = B - 1.

PRINT ('Parked at B', B)

Bike + 1, Accepted + 1.

ELSE: Rejected + 1.

PRINT ('Vehicle Rejected').

ENDIF.

IF type = 'V':

~~IF B > 0:~~ IF B > 1:

~~B = B - 1.~~ B = B - 2

PRINT ('Vehicle Parked at B', B)

Accepted + 1, Van + 1.

ELSE: Rejected + 1.

PRINT ('Vehicle Rejected').

ENDIF

ENDIF.

IF Zone = 'C':

IF type = 'A':

IF C > 0:

PRINT ('Parked at C', C)

C = C - 1

Car + 1, Accepted + 1

```
ELSE = Rejected += 1.  
PRINT('Parked at C', C)  
ENDIF.
```

```
IF type = 'B':  
    IF C > 0:  
        C = C - 1.  
        PRINT('Parked at C', C)  
        Accepted += 1, Bike += 1  
    ELSE:  
        PRINT('Vehicle Rejected')  
        Rejected += 1.  
ENDIF
```

```
IF type = 'V':  
    IF C > 1:  
        C = C - 2.  
        PRINT('Parked at C', C)  
        Accepted += 1, Van += 1.  
    ELSE:  
        PRINT('Vehicle Rejected')  
        Rejected += 1.  
ENDIF  
ENDIF
```

Vehicle = Vehicle - 1.
ENDWHILE.

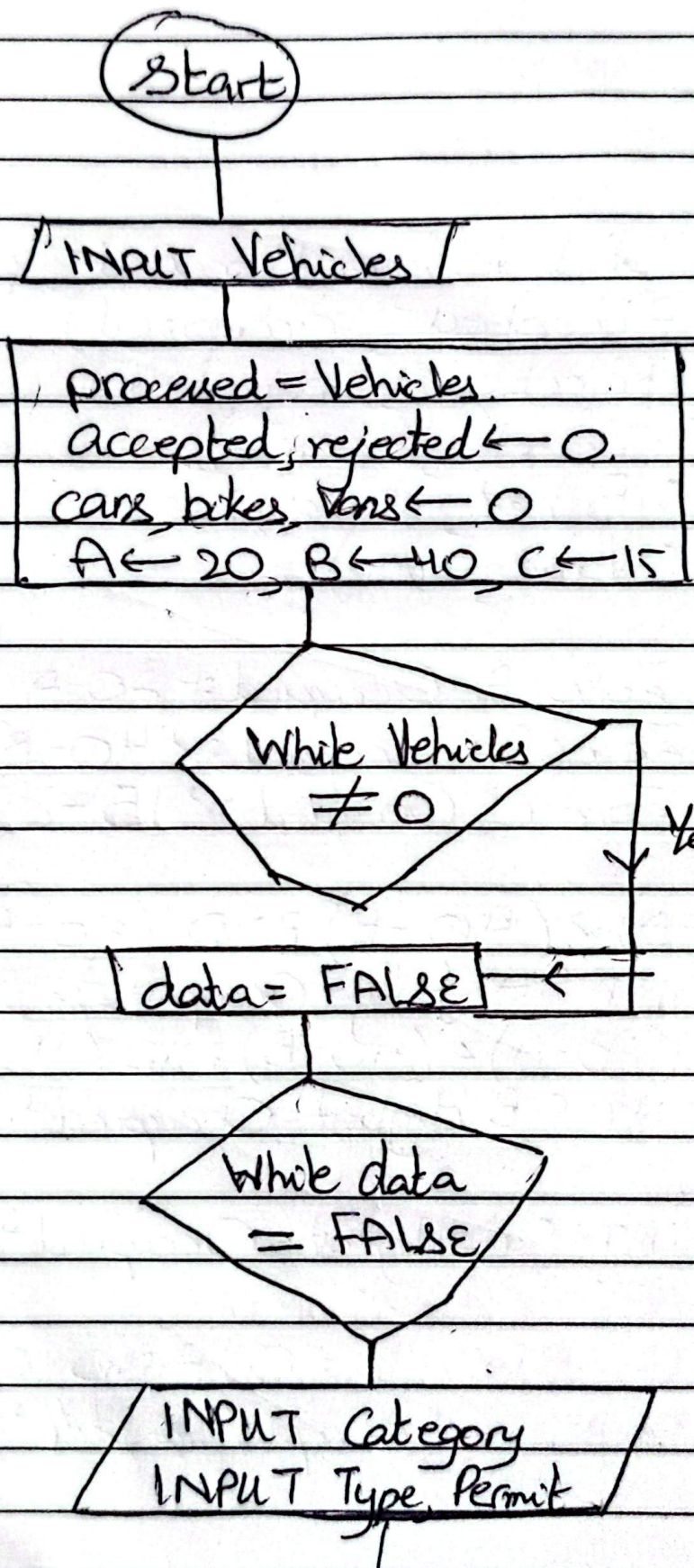
PRINT('Total Vehicles Processed', processed).
PRINT('Accepted', accepted).
PRINT('Rejected', rejected).
PRINT('Cars', cars).
PRINT('Bikes', bikes).
PRINT('Van', Van).

PRINT('Zone A Occupied:', 20-A, 'left', A).
PRINT('Zone B Occupied:', 40-B, 'left', B).
PRINT('Zone C Occupied:', 15-C, 'left', C).

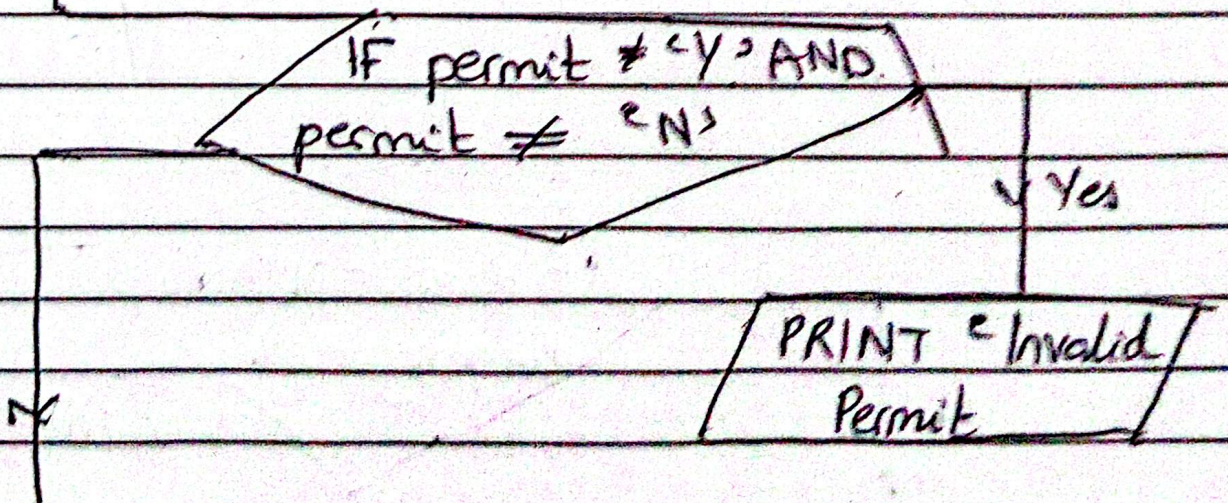
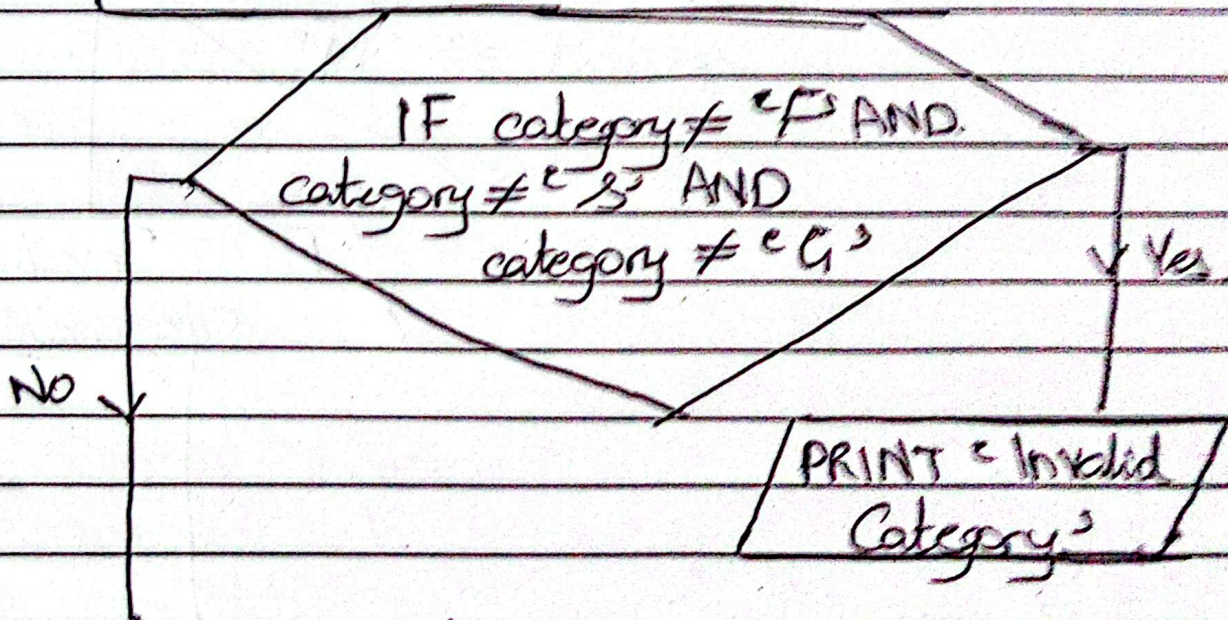
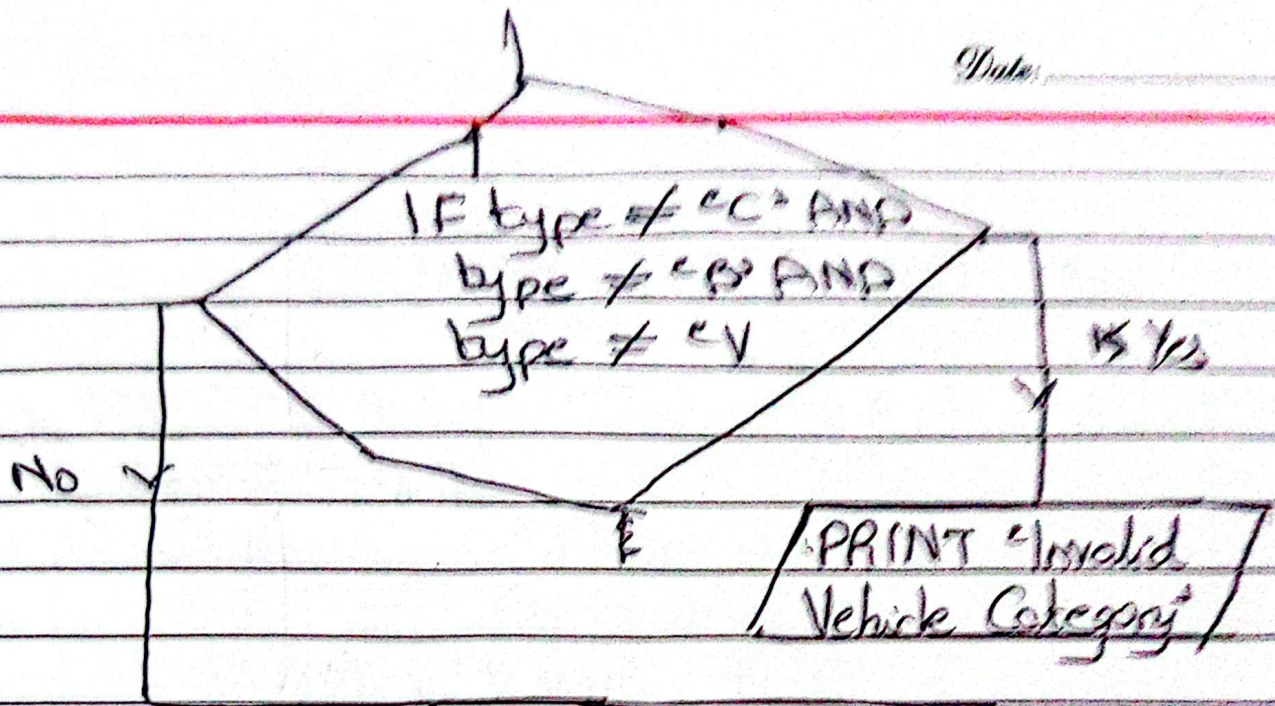
IF (20-A) > (40-B) AND (20-A) > (15-C):
 PRINT 'A fully Occupied'.
ELSE IF (20-A) < (40-B) AND (40-B) > (15-C):
 PRINT 'B Highest Occupied'.
ELSE.

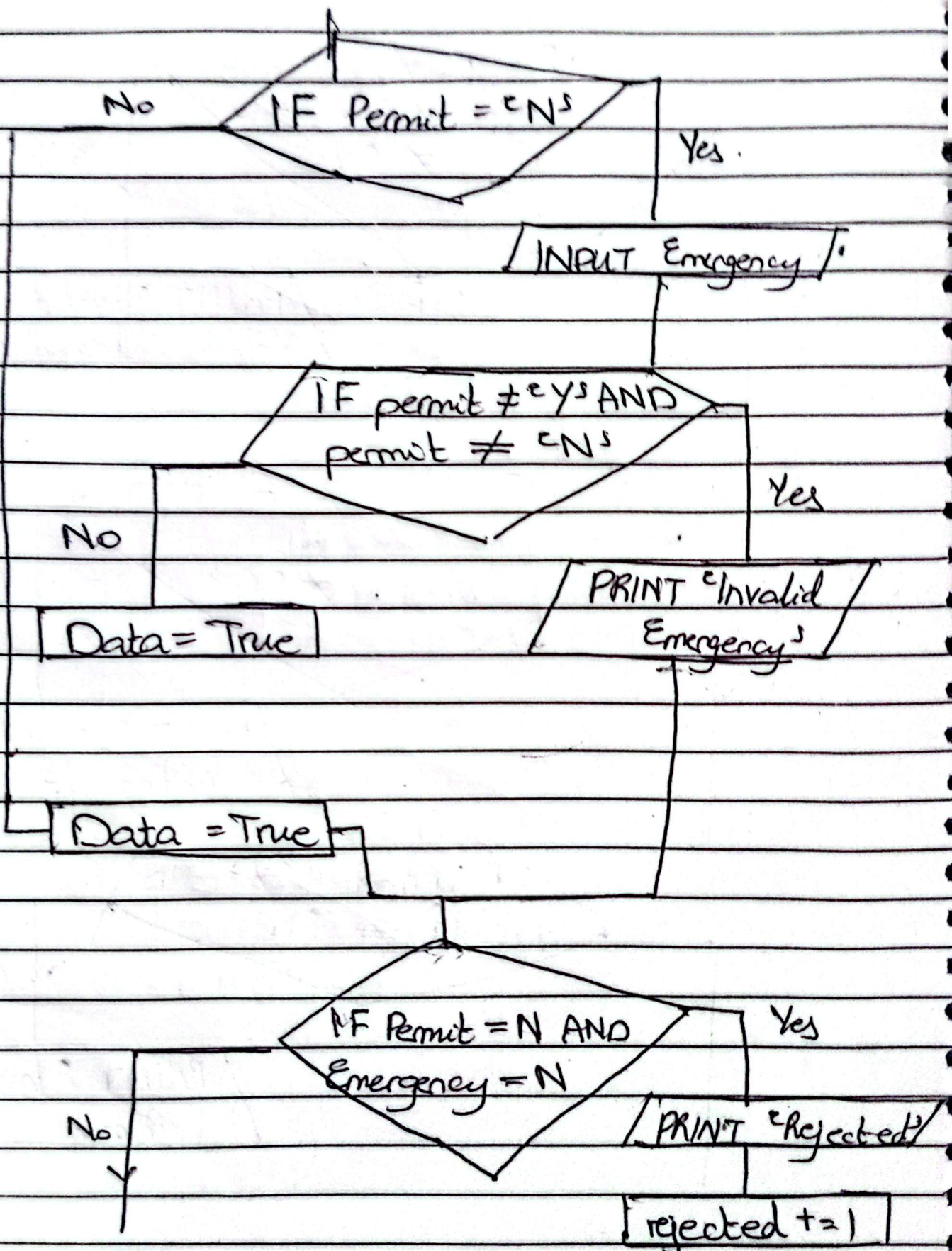
 PRINT 'C Highest Occupied'.

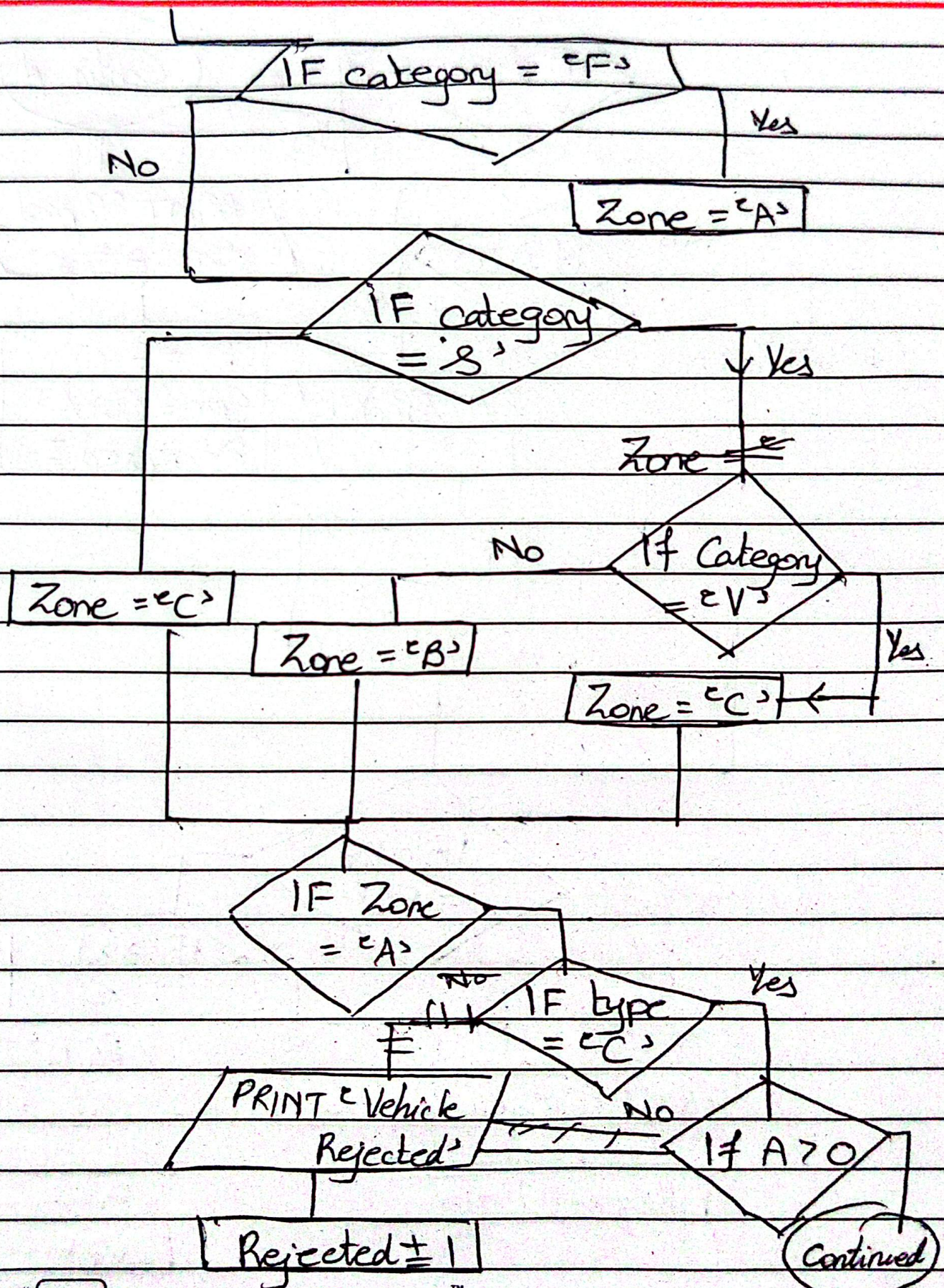
IF A = 0 AND B = 0 AND C = 0
 PRINT('Campus is full').
ENDIF

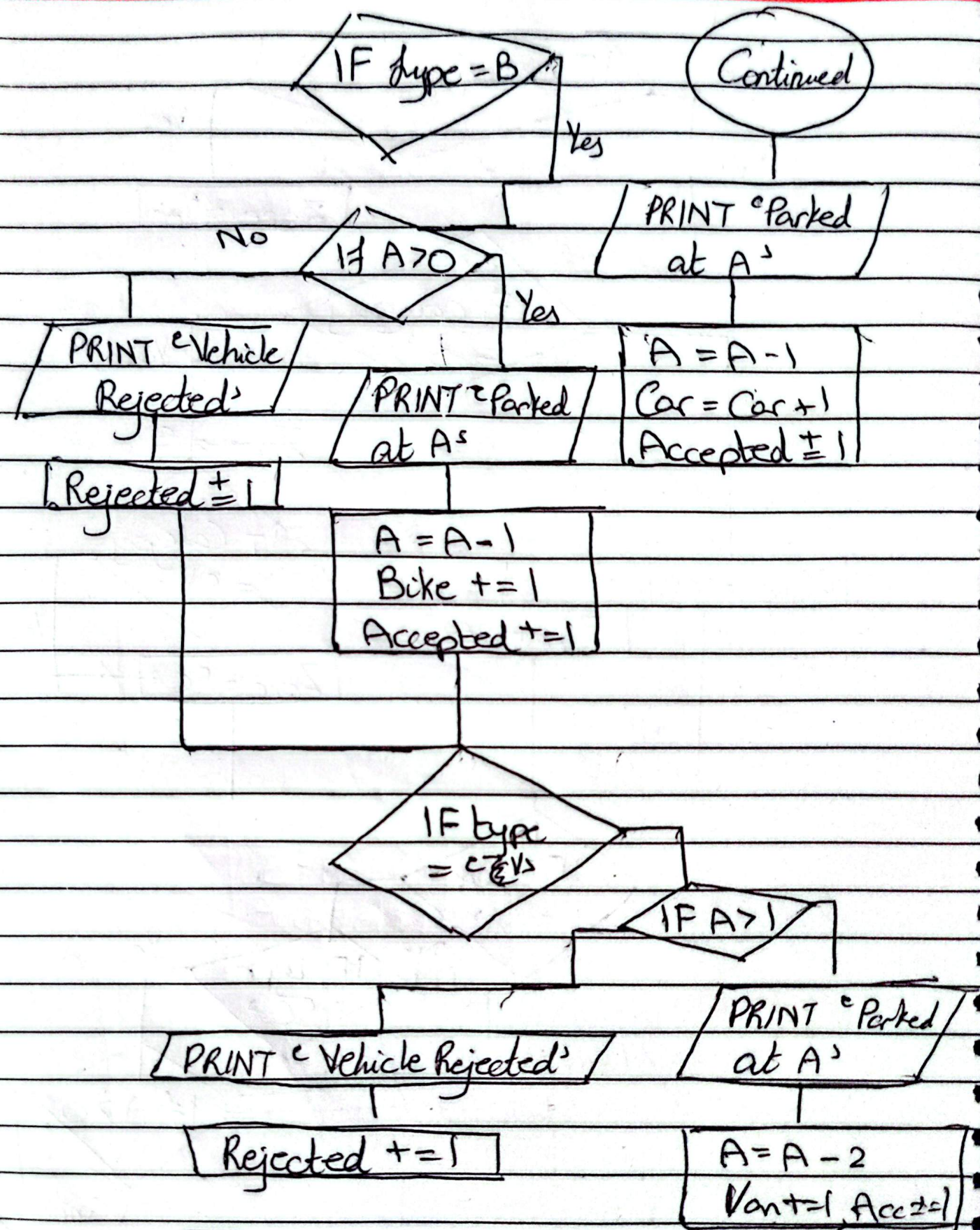


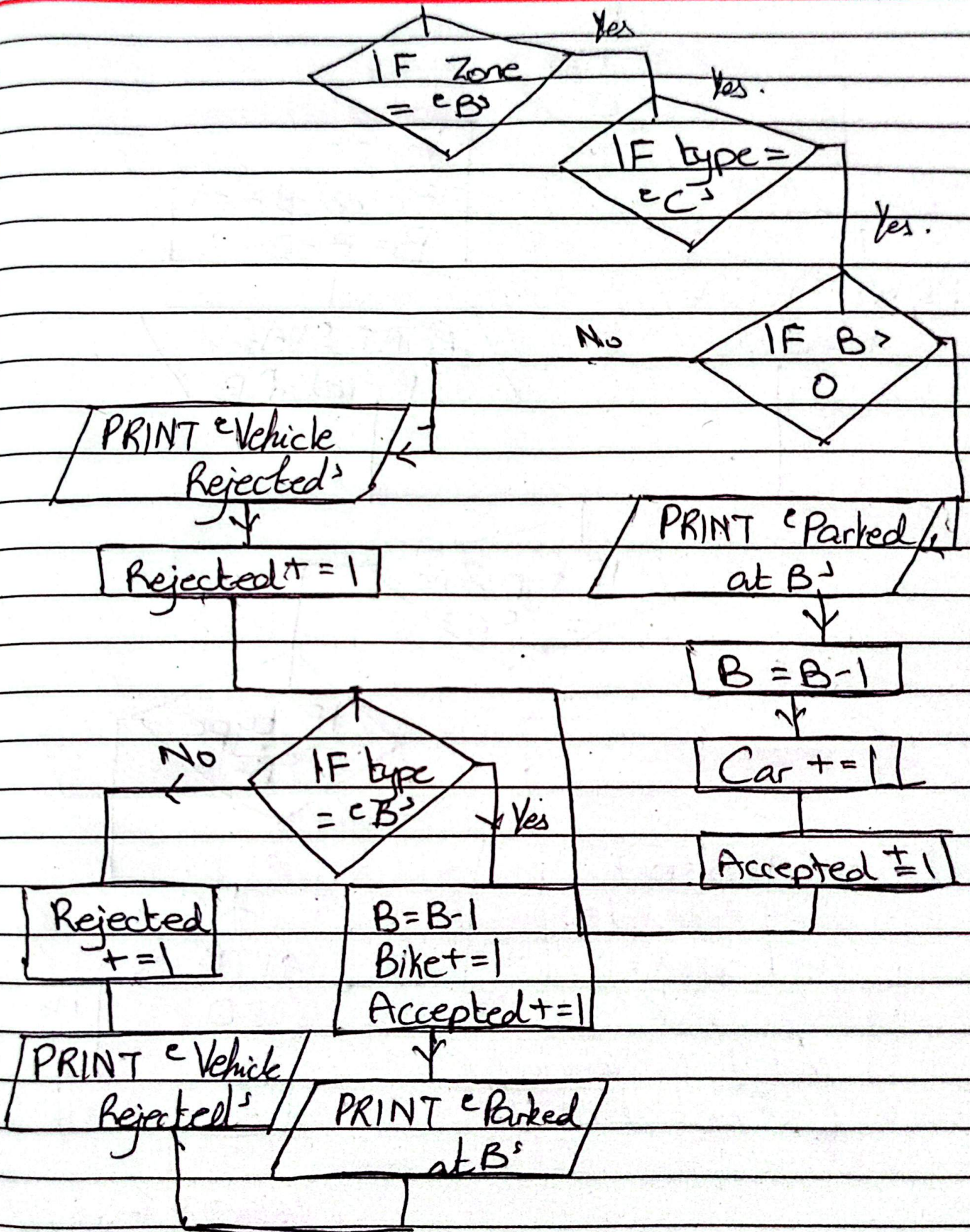
Date: _____

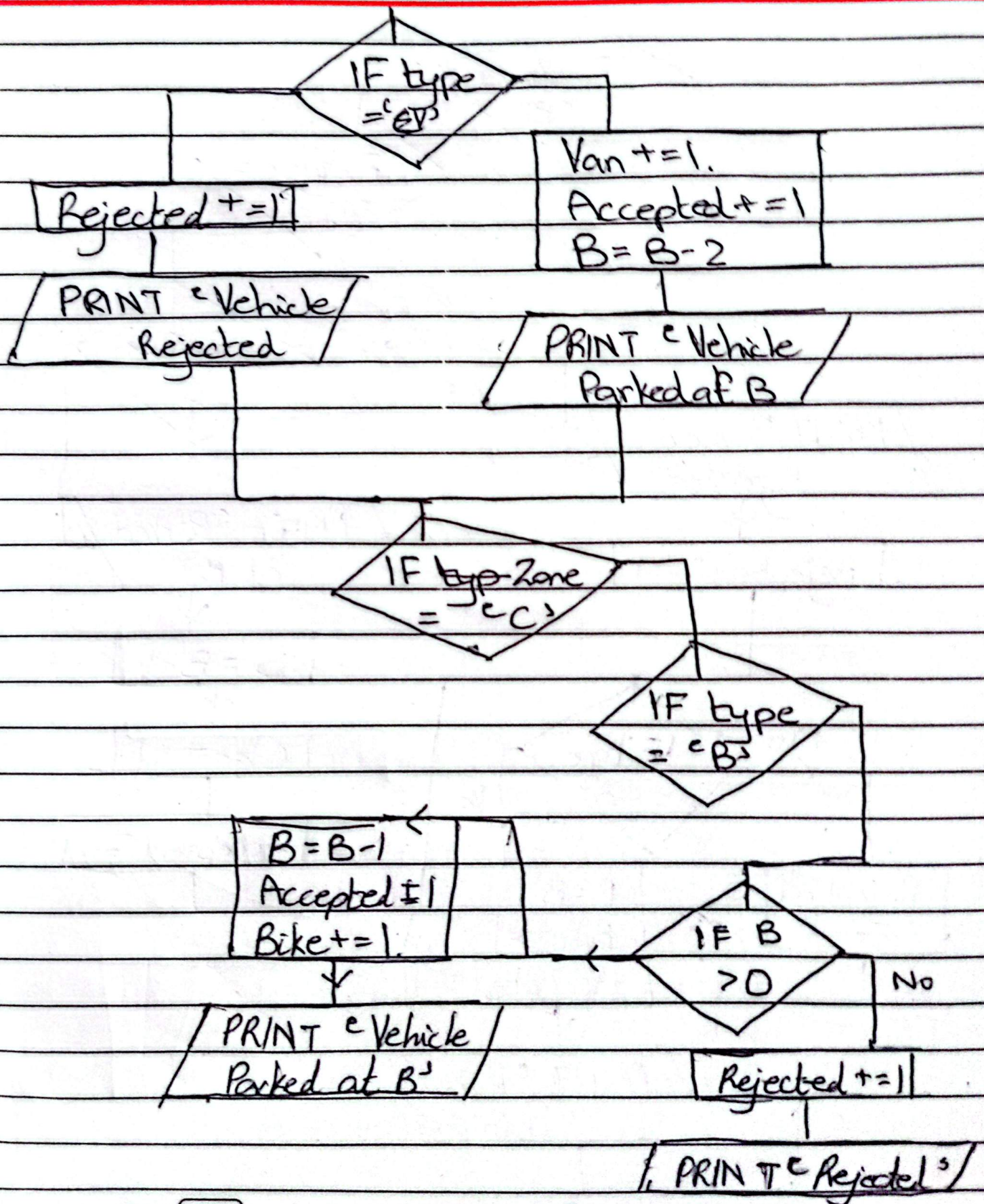


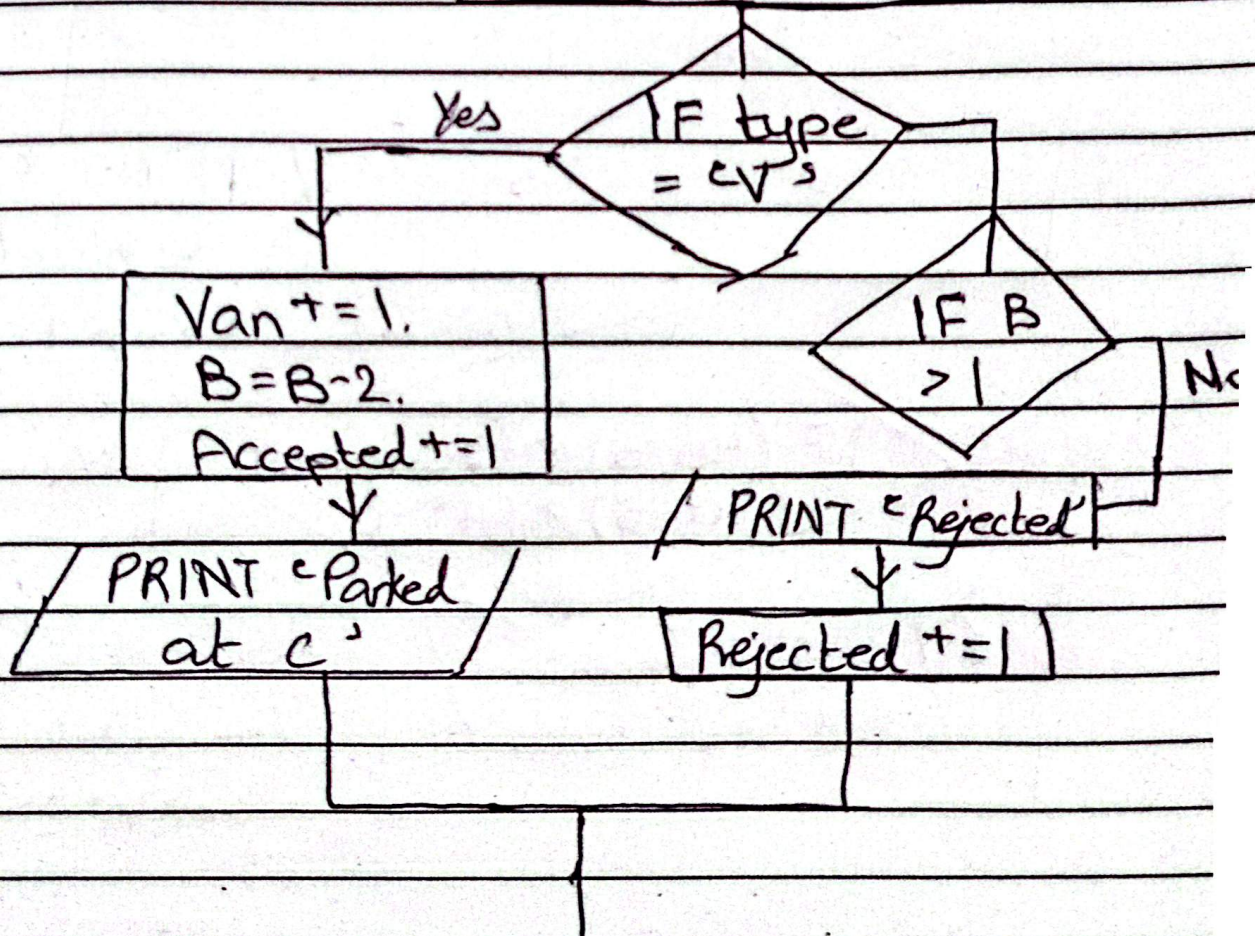
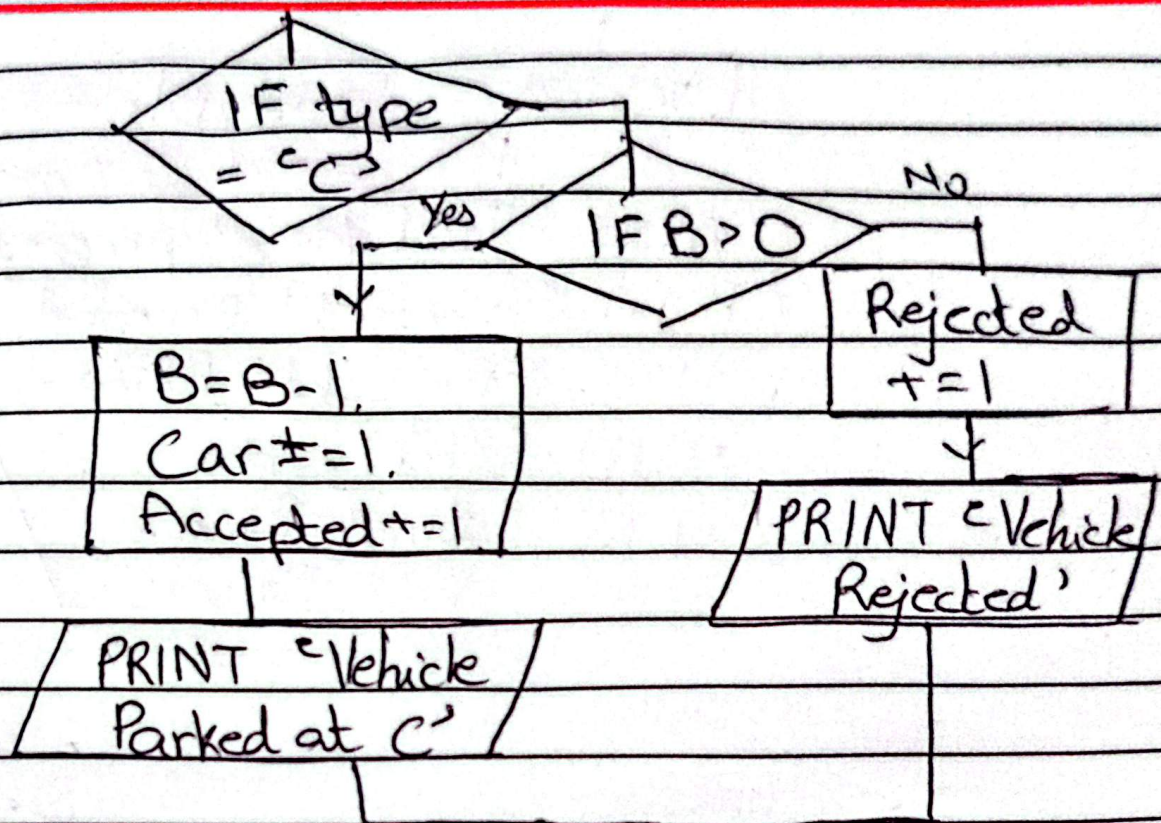


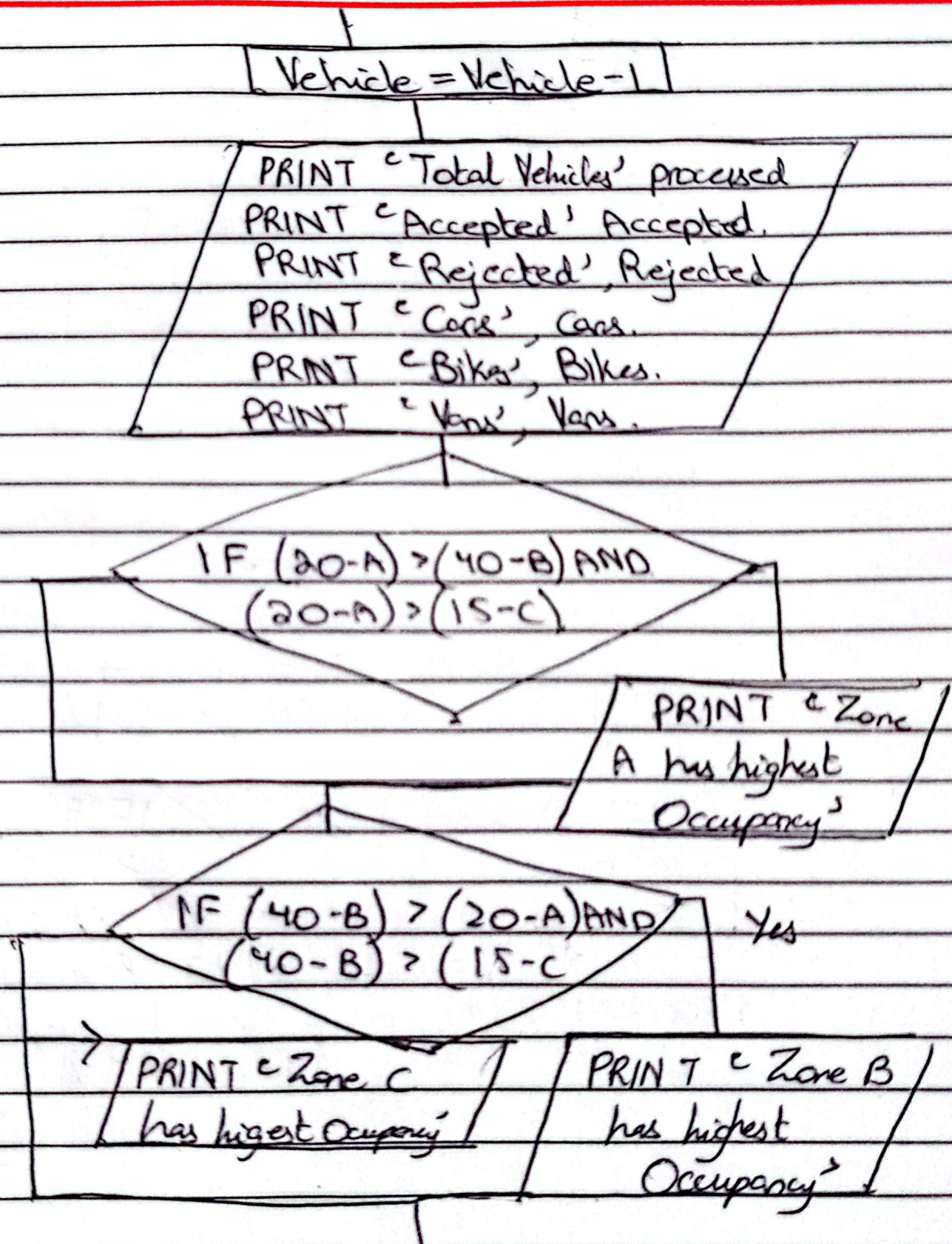












Date: _____

